☑ High Priority

CONTRACTOR SUBMITTAL TRANSMITTAL FORM REV. A

Owner: County of Hawaii
Contractor: Nan, Inc. **Project No.:** WW-4705R
Project Name: Hilo WWTP Phase 1 **Submittal Number:** 06611-001.0
Submittal Title: FRP Fabrications - Pultruded Grating ☐ For Information Only
TO: County
From: Nan Inc.

Specification No. and Subject of Submittal / Equipment Supplier			
Spec: 06611	**Paragraph:** 1.03 A & B, 2.05		
Authored By: Fibergrate		**Date Submitted:** 09/30/25	

Submittal Certification

Check Either (A) or (B):

☑ (A) We have verified that the equipment or material contained in this submittal meets all the requirements specified in the project manual or shown on the contract drawings with no exceptions.

☐ (B) We have verified that the equipment or material contained in this submittal meets all the requirements specified in the project manual or shown on the contract drawings except for the deviations listed.

Certification Statement: By this submittal, I hereby represent that I have determined and verified all field measurements, field construction criteria, materials, dimensions, catalog numbers and similar data, and I have checked and coordinated each item with other applicable approved shop drawings and all Contract requirements.

General Contractor's Reviewer's Signature: Marshall Rucknagel

Printed Name and Title: Marshall Rucknagel, Project Engineer

In the event, Contractor believes the Submittal response does or will cause a change to the requirements of the Contract, Contractor shall immediately give written notice stating that Contractor considers the response to be a Change Order.

Firm: **Signature:** **Date Returned:**

PM/CM Office Use

Date Received GC to PM/CM:
Date Received PM/CM to Reviewer:
Date Received Reviewer to PM/CM:
Date Sent PM/CM to GC:

Nan, Inc

PROJECT: HILO WWTP REHABILITATION AND REPLACEMENT PROJECT - PHASE 1

JOB NO. WW-4705R

THIS SUBMITTAL HAS BEEN CHECKED BY THIS CONTRACTOR. IT IS CERTIFIED CORRECT, COMPLETE, AND IN COMPLIANCE WITH CONTRACT DRAWINGS AND SPECIFICATIONS. ALL AFFECTED CONTRACTORS AND SUPPLIERS ARE AWARE OF, AND WILL INTEGRATE THIS SUBMITTAL (UPON APPROVAL) INTO THEIR OWN WORK.

DATE RECEIVED 9/30/2025
SPECIFICATION SECTION # 06611
SPECIFICATION Fiberglass Reinforced Plastic
PARAGRAPH 1.03, 2.05
DRAWING Fibergate
SUBCONTRACTOR na
SUPPLIER na
MANUFACTURER na

CERTIFIED BY CQCM or Designee :

SECTION 06611

FIBERGLASS REINFORCED PLASTIC FABRICATIONS

PART 1 GENERAL

1.01 SUMMARY

A. Section includes: Fiberglass reinforced plastic fabrications including:
   1. Weirs.
   2. Baffles.
   3. Parshall flume liners and Cutthroat flume liners.
   4. Molded grating.
   5. Pultruded grating.
   6. Tank covers.
   7. Troughs.

1.02 REFERENCES

A. American Water Works Association (AWWA):
   1. F102 - Matched-Die-Molded, Fiberglass-Reinforced Plastic Weir Plates, Scum Baffles, and Mounting Brackets.

B. ASTM International (ASTM):
   1. D635 - Standard Test Method for Rate of Burning and/or Extent and Time of Burning of Plastics in a Horizontal Position.
   2. D638 - Standard Test Method for Tensile Properties of Plastics.
   3. D695 - Standard Test Method for Compressive Strength of Plastics.
   4. D790 - Standard Test Methods for Flexural Properties of Unreinforced and Reinforced Plastics and Electrical Insulating Materials.
   5. D2583 - Standard Test Method for Indentation of Hardness of Rigid Plastics by Means of a Barcol Impressor.
   6. E84 - Standard Test Method for Surface Burning Characteristics of Building Materials.

C. United States, Department of Agriculture (USDA).

D. United States, Department of the Interior:
   1. Bureau of Reclamation (USBR):
      a. ISO 9826 - Water Measurement Manual, Measurement of Liquid Flow in Open Channel.

1.03 SUBMITTALS

A. Shop Drawings: Plans and elevations showing sizes, shapes, locations of installations, and details of connections and anchorage.

B. Submit general arrangement and fabrication drawings, calculations, and elements of the design for grating and stairs.

C. Product Data and Material Certifications.

D. Manufacturer's Installation Instructions.

E. For tank covers:
   1. Submit structural design calculations prepared and signed by a professional structural engineer licensed in the state of Hawaii.

1.04 DELIVERY, STORAGE, AND HANDLING

A. Parshall flume liners and Cutthroat flume liners: Provide temporary bracing for liner to ensure maintenance of dimensions during shipment. Maintain bracing in place for installation.

B. Tank covers: Meet the requirements of the manufacturer’s written instructions. Handle and store cover system components using equipment and components that will prevent puncture, scratching, and abrasion of fiberglass laminates.

PART 2 PRODUCTS

2.01 WEIR PLATES

A. General:
   1. In accordance with AWWA F102, and as specified in this Section.

B. Materials:
   1. Laminate construction: Glass fiber-reinforced, chemical-resistant polyester resin.
      a. Glass content of laminate: Minimum 20 percent by weight.
      b. Resin fillers: Minimum 40 percent.
   2. Physical properties of laminate: Equal to or exceeding the tensile, flexural, impact, hardness, and water absorption properties specified in AWWA F102.
   3. Performance criteria:
      a. Chemical resistance: Classification: AWWA F102 Type II (chemical-resistant resin, filler, and fabrications)
   4. Color: Manufacturer’s standard aqua/turquoise.

C. Fabrications:
   1. Fabricated to the shapes, dimensions, and details indicated on the Drawings and specified, and using processes as specified in this Section.
   2. Dimensions:
      a. Conform to tolerances in accordance with AWWA F102, unless otherwise indicated.
      b. Final laminate thickness: Plus or minus 10 percent of nominal thickness.
   3. Fabricate weir plates with oversize holes to allow vertical adjustment.
   4. Seal cut edges of non-standard lengths, and edges of drilled and countersunk holes in fiberglass reinforced plastic fabrications with resin.
   5. Furnish fiberglass reinforced plastic lapped plate splices at joints.

D. Rectangular "flat crested" weirs:
   1. Fabrication:
      a. Molded to produce uniform, smooth surfaces.
      b. Tolerances: In accordance with AWWA F102.
   2. Manufacturers: One of the following or equal:
      a. Glass Steel, Inc.
      b. MFG Water Treatment Products.
      c. NEFCO, Inc.
      d. Warminster Fiberglass.

E. Accessories:
   1. Washers: Of same material as weirs, with surfaces smooth, free of voids, and without dry spots and crazes.
   2. Assembly hardware: Concrete anchors as specified in Section 05190 - Mechanical Anchoring and Fastening to Concrete and Masonry.

2.02 WEIR PLATES WITH SCUM BAFFLES

A. Manufacturers: One of the following or equal:
   1. MFG Water Treatment Products.
   2. Warminster Fiberglass Company.

B. Materials:
   1. Match die molded construction.
   2. Minimum Corrosion Liner:
      a. One "C" or Nexus veil as specified for the service environment.
      b. Remainder 1-1/2 ounce per square foot mat to total minimum thickness of 0.096 inches on surface exposed to the service environment.
   3. Ultraviolet Stabilizer: Added to the exterior surface coat of fabrications intended for outside service, in the type and amount recommended by the resin manufacturer.
   4. Resin: Premium grade vinyl ester, manufactured by one of the following or equal as recommended by the resin manufacturer for the specific operating environment:
      a. Dow Chemical Company, Derakane 411.
      b. Ashland Chemical Company, Hetron 922.
      c. Reichhold Inc., Reichhold Dion VER 9100.
      d. Interplastic Corporation, Interplastic VE 8300.
   5. Color: Natural, unless otherwise specified.

C. Fabrication:
   1. Scum Baffles: Match die molded components to specified shape and dimensions.
   2. Weirs:
      a. Compression molded in matched metal die molds. Fabrications from plate stock with cut-in edges, ends, or V-notches will not be accepted.
      b. Tolerances: In accordance with AWWA F 102, except as follows:
         1) Deviation between top elevations of V-notches in a line: Maximum 1/32 inch in 6-foot length of weir plate.

2) Deviations between bottom elevations of V-notches in a line: Maximum 1/32 inch in 6-foot length of weir plate.
3) Deviations in depth of V-notches in a line: Maximum 1/32 inch in 6-foot length of weir plate.

c. Weir Plates:
1) Of shape and dimensions specified.
2) Provide 2-3/8 inch diameter holes for adjustment.
3) Resin coat cut edges, and drilled and countersunk holes in fiberglass reinforced plastic fabrications.

d. Furnish fiberglass reinforced plastic butt plates for joints.

D. Design Criteria and Chemical Exposure: Suitable for installation in a secondary clarifier in a municipal wastewater treatment plant.

E. Accessories:
1. Washers: Of same material as weirs, with surfaces smooth, free of voids, and without dry spots and crazes.
2. Mounting hardware shall be 316 stainless steel bolt type fasteners.[AD8]

~~2.02~~2.03 PARSHALL FLUME LINER AND CUTTHROAT FLUME LINER

A. General:
1. Size(s)/Dimensions: As indicated on the Drawings with interior dimensions in accordance with USBR/ISO 9826.
2. Performance requirements:
   a. Accuracy of flow: Plus or minus 5 percent of rate with plus or minus 0.5 percent of rate repeatability.
   b. Chemical exposure.
   c. Suitable for outdoor use in contact with raw and screened sewage at temperatures ranging from 50 to 85 degrees Fahrenheit.

B. Manufacturers: One of the following or equal:
1. Plasti-Fab Inc.
2. TRACOM, Inc.
3. Warminster Fiberglass.

C. Parshall flumes and Cutthroat flumes (concrete):
1. Composition: Sufficient embedded galvanized steel to produce substantial, self-supporting rigid structure requiring no external supporting structure.
2. Cutthroat flumes at the headworks:
   a. Throat width (with fiberglass reinforced plastic liner installed): 12 inches.
   b. Flume length: 108 inches.
3. Interior dimensions with fiberglass reinforced plastic liner installed: In accordance with USDA Circular 843.a.

D. Materials:
   1. Glass fiber reinforced plastic, having the following properties:

Test	Standard	Requirement
Tensile strength	ASTM D638	14,000 psi, minimum
Flexural strength	ASTM D790	25,000 psi, minimum
Flexural modulus	ASTM D790	1,000,000 psi, minimum
Indentation hardness (Barcol)	ASTM D2583	40 minimum, average

   2. Minimum corrosion liner:
      a. On interior surface of flume.
      b. 2 "C" or Nexus veils as specified for the service environment.
      c. Remainder 1-1/2 ounce per square foot mat to a total minimum thickness of 0.106 inch.
   3. Ultraviolet stabilizer: Added to the corrosion liner in the type and amount recommended by the resin manufacturer.
   4. Minimum 3/8-inch total thickness.
   5. Resin: Premium grade vinyl ester:
      a. Manufacturers: One of the following or equal: As recommended by the resin manufacturer for the specific operating environment:
         1) Derakane 411.
         2) Hetron 922.
         3) Reichhold Dion VER 9100.
         4) Interplastic VE 8300.
   6. Color: Natural, unless otherwise specified.

E. Fabrication:
   1. Fabricated with integral stiffening ribs.
   2. One piece, full-length construction for flumes up to 84 inches.

~~2.03~~2.04MOLDED GRATING

A. Manufacturers: The following or approved equal:
   1. Fibergrate Composite Structures, Inc., Dallas, TX:
      a. Fibergrate Molded Grating, Square Mesh.

B. Materials:
   1. Core: Bi-directionally aligned glass fibers.
   2. Mat: Submit standard corrosion liner.
   3. Veil: Submit standard corrosion liner.
   4. Ultraviolet stabilizer: Added to the exterior surfaces in the type and amount recommended by the resin manufacturer.
   5. Resin: Fire retardant premium grade corrosion-resistant vinyl ester, antimony trioxide or pentoxide added to meet Class I flame spread rating of ASTM E84 and self-extinguishing requirements of ASTM D635. Resin containing fillers or pigments of any kind is prohibited.
   6. Manufacturer: One of the following as recommended by the resin manufacturer for the specific operating environment:
      a. Dow Derakane 411.

        b. Ashland Hetron 922.
    7. Color: gray.
    8. Anti-slip coating: Grating shall be manufactured with an integrally applied grit to the top surface of each bar providing maximum slip resistance.

C. Design criteria and chemical exposure:
    1. Depth: 2 inches with tolerance of ±1/16 inch.
    2. Mesh configuration: 2 inches by 2 inches with tolerance of ±1/16 inch centerline to centerline.
    3. Panel size: 4 feet by 12 feet standard panel sizes.
    4. Grating bar intersections filleted to minimum radius of 1/16 inch.
    5. Deflection and concentrated load: Maximum 0.04 inches at span of 30 inches under concentrated load of 300 pounds.
    6. Deflection and uniform distributed load: Capable of carrying uniform distributed load of 100 pounds per square foot on simple span of 43 inches without deflecting more than 0.25 inches.
    7. Manufacturer to certify stiffness of all panels never more than 2.5 percent below published load-deflection values.
    8. Weight: Maximum 4.0 pounds per square foot for 2-inch high grating with open area of minimum 50 percent.
    9. Suitability: Use grating suitable for use in environments containing sulfuric acid solutions at the temperatures and concentrations specified for the application. Grating is intended for immersion surface. Manufacturer shall verify that all finished surfaces shall be smooth, resin-rich, free of voids and without dry spots, cracks, crazes or unreinforced areas. All glass fibers shall be well covered with resin to protect against their exposure due to wear or weathering.

D. Components:
    1. Grating shall be one-piece molded construction with tops and bottoms of rectangular bearing bars in the same plane arranged in a regular square grid pattern.
    2. Hold downs, connectors, and accessories: Type 316 stainless steel.

E. Fabrication:
    1. Produce grating as one piece molded construction.
    2. Produce grating with square mesh pattern.
    3. Reinforce with continuous rovings of equal number of layers in each direction.
    4. Maximum 1/8-inch clearance between top layer of reinforcement and top surface of grating.
    5. Coat ends of grating with resin.
    6. Fabricate single sections for each span. Do not clamp 2 or more grating sections together within spans.
    7. Type 316 stainless steel plates and angles at openings:
        a. Install 5/8-inch thick plate or angle where required to fill openings at changes in elevation and at openings between equipment and grating.
        b. Install angle stops at ends of grating to prevent grating from sliding.
    8. Maximum 1/8-inch clearance allowed between ends of grating and inside face of vertical leg of support angles.
    9. Percentage of glass shall not exceed 35 percent by weight.
    10. Following molding, no dry glass fibers shall be visible on any surface of bearing bars or cross bars. All bars shall be smooth and uniform with no

evidence of fiber orientation irregularities, interlaminar voids, porosity, resin-rich, or resin-starved areas.

F. Cutouts:
  1. Provide where required for equipment access or penetrations, including valve operators, stems, and gate frames.
  2. Seal cut edges with resin.

~~2.04~~2.05PULTRUDED GRATING

A. Manufacturers: One of the following or approved equal:
  1. Fibergrate Composite Structures, Inc., Dallas, TX:
    a. Fibergrate Safe-T-Span, Pultruded Grating.

B. Materials:
  1. Fiberglass reinforcement: Shall be a combination of continuous roving, continuous strand mat, and surfacing veil in sufficient quantities as needed by the application and/or physical properties required.
  2. Grating components: Shall be high strength and high stiffness pultruded elements having a maximum of 70 percent and a minimum of 60 percent glass content (by weight) of continuous roving and continuous strand mat fiberglass reinforcements. The finished surface of the product shall be provided with a surfacing veil to provide a resin rich surface, which improves corrosion resistance and resistance to ultraviolet degradation. Bearing bars shall be interlocked and epoxied in place with a two-piece cross rod system to provide a mechanical and chemical lock. Cross rods should be below the walking surface of the grating. Gratings with cross rods that are flush with the walking surface are excluded.
  3. Veil: Submit standard corrosion liner.
  4. Ultraviolet stabilizer: Added to the exterior surfaces in the type and amount recommended by the resin manufacturer.
  5. Resin: Fire retardant premium grade corrosion-resistant vinyl ester, antimony trioxide or pentoxide added to meet Class I flame spread rating of ASTM E84 and self-extinguishing requirements of ASTM D635. Resin containing fillers or pigments of any kind is prohibited:
    a. Manufacturer: As recommended by the resin manufacturer for the specific operating environment.
  6. Color: Gray.
  7. Anti-slip coating: Grating shall be provided with a quartz grit bonded and baked to the top surface of the finished grating product.

C. Design criteria and chemical exposure:
  1. Depth: 1.5-inches with tolerance of ±1/32 inch.
  2. Configuration: 1.5-inch load bar spacing with 6-inch tie bar spacing on centers.
  3. Span: As indicated on the Drawings.
  4. Deflection and concentrated load: Maximum 0.11 inches at span of 48 inches under concentrated load of 300 pounds.
  5. Deflection and uniform distributed load: Capable of carrying uniform distributed load of 100 pounds per square foot on simple span of 48 inches without deflecting more than 0.08 inches.

6. Manufacturer to certify stiffness of all panels never more than 2.5 percent below published load-deflection values.
7. Weight: Maximum 4.0 pounds per square foot for 2-inch high grating with open area of minimum 50 percent.
8. Suitability: Use grating suitable for use in environments containing sulfuric acid solutions at the temperatures and concentrations specified for the application. Grating is intended for immersion surface. Manufacturer shall verify that all finished surfaces shall be smooth, resin-rich, free of voids and without dry spots, cracks, crazes or unreinforced areas. All glass fibers shall be well covered with resin to protect against their exposure due to wear or weathering.

D. Components:
1. Grating shall be pultruded construction with tops and bottoms of rectangular bearing bars in the same plane arranged in a regular pattern with intermittent cross bars for lateral stability bracing.
2. Hold downs, connectors, and accessories: Type 316 stainless steel.

E. Fabrication:
1. Measurements: Grating supplied shall meet the minimum dimensional requirements as shown or specified. The Contractor shall provide and/or verify measurements in field for work fabricated to fit field conditions as required by grating manufacturer to complete the work. Determine correct size and locations of required holes or cutouts from field dimensions before grating fabrication.
2. Layout: Each grating section shall be readily removable, except where indicated on the Drawings. The manufacturer shall provide openings and holes where indicated on the Drawings. Supplemental supports/stiffeners shall be provided at openings in the grating by the manufacturer where necessary to meet load/deflection requirements specified herein. Grating openings shall be fabricated at the edge of individual panels as required to allow for the grating sections to be removed.
3. Sealing: All shop fabricated grating cuts shall be coated with vinyl ester resin to provide maximum corrosion resistance. All field fabricated grating cuts shall be coated similarly by the Contractor in accordance with the manufacturer's recommendations.

~~2.05~~2.06 TANK COVERS

A. General:
1. This section includes specifications for:
   a. Tank cover deck panels.
   b. Structural supports.
   c. Flashing and trim.
   d. Fasteners and anchors.
   e. Bulb gaskets and sealant.
   f. Accessories and appurtenances.

B. Manufacturers: One of the following or equal:
1. Enduro Composites.
2. NEFCO, Inc.

C. Materials:

1. FRP Physical and mechanical properties:

Method	Test	Minimum Value
ASTM D638	Tensile Strength	40,000 psi
ASTM D790	Flexural Strength	45,000 psi
ASTM D790	Flexural Modulus	2.27 x 10^6 psi
ASTM D256	Notched Izod Impact	20 ft-lbs/in
ASTM D570	Water Absorption	0.20 percent

2. Resin:
   a. Industrial quality, fire retardant isophthalic resin.
   b. Pigmented to ensure that the resulting component is opaque.
   c. Containing UV stabilizers for protection against sunlight.
3. Cover Panels:
   a. Cover panels shall be sealed at side-laps with non-adhesive, 1” diameter neoprene bulb gasket per ASTM C864. Side lap gaskets shall be factory installed and oriented vertically to be compressed when adjacent panel is placed into position.
   b. Cover panels shall meet the following criteria:
      1) The cover shall have a slip-resistant walking surface designed for operator foot traffic.
      2) Non-skid surface shall be an integral (embedded) part of the cover panel and not achieved by use of coating/paint, adhesive tapes, or other applied product after the manufacturing process.
      3) Slip resistance tread shall be bi-directional.
      4) Minimum average Dynamic Coefficient of Friction (DCOF) shall be 0.50 per ANSI A137.1/A326.3 Dynamic Coefficient of Friction Test.
      5) 3.Minimum wet Pendulum Test Value (PTV) shall be 45, with Four S (96) hard rubber slider, per AS HB198:2014 (AS/NZS 4586) Pendulum Test.
4. Glass reinforcement:
   a. Continuous strand rovings and mat 50 percent by weight.
5. All surfaces shall have a UV resistant factory applied non-skid tape.
6. Hardware:
   a. All fasteners, anchorage, hinges, handles, latches, brackets, and other structural accessories shall be provided by the cover manufacturer.
   b. Fasteners, handles, latches, perimeter flashing anchors, concrete anchors and other hardware shall be 316 Stainless Steel.
7. Access Hatches and View Ports:
   a. Flat view port hatches shall be 12 inches square or less.
   b. Hatch openings shall be factory cut in cover decking panel by manufacturer.
   c. Hatch lids shall have factory applied nonskid UV resistant surface with stainless steel lift handles.
   d. The underside of hatch lids shall be sealed with factory installed neoprene bulb gasket. The perimeter hatch curb shall be sealed to decking surface with adhesive sealant.

e. Hatches shall have stainless steel hold-open device to prevent door from blowing open or closing on itself. Hatches shall be secured with hand-operable latches and without special tools.
8. Air Vents and Connections:
   a. FRP gooseneck ventilation (as indicated on drawings) with bird screen shall be provided by cover manufacturer.
9. Gaskets and Sealants:
   a. All panel side laps, and perimeter conditions shall be gasketed.
   b. Gaskets located at panel side laps shall be factory installed, 1-inch diameter neoprene, bulb type.
   c. Gaskets under non-radius end flashing shall be factory installed, bulb type.
   d. Adhesive sealant shall be applied by contractor at various locations as required by manufacturer for odor containment.

D. Design criteria:
1. Covers shall be designed and manufactured to inhibit to the extent possible incident sunlight from striking the surfaces of the launder and weir.
2. Neither the cover nor the means to support it shall interfere with effluent flow over the weirs or within the trough.
3. Cover will be supported on the concrete curb.
4. All joints at the wall will have a sponge neoprene gasket between the angle and the cover.
5. There will also be a sponge neoprene gasket radially between each cover panel.
6. Panel requirements:
   a. Panel Dimensions as designed by manufacturer.
   b. Flat and walkable with a non-skid surface on the walking area.
   c. Designed for a 50 pounds per square foot pedestrian, or 500 lbs concentrated load on an area 30 inches by 30 inches (whichever has greater effect) with a maximum deflection of L/240.
   d. Designed for a wind uplift pressure of 56 psf.
   e. Design structural beam underneath the panel. All structural components of the beams shall be 316 Stainless Steel.
7. Air Leakage:
   a. Air leakage shall not exceed 1 CFM/LF at gasketed panel joints and 2.2 CFM/LF at cover perimeter under -.5 inch water pressure per HVAC Air Duct Leakage Test in accordance with NEBB "Procedural Standards for Adjusting, Balancing, of Environmental Systems".
   b. Compliance shall be certified by approved National Environmental Balancing Bureau (NEBB) agency.
8. Cover Panel Removability:
   a. Each cover panel shall be removable without having to remove no more than its two, adjacent panels.
   b. Each cover panel shall be removable vertically and without cutting of a cover component.

~~2.06~~2.07 TROUGHS

A. Manufacturers: One of the following or equal:
1. Warminster Fiberglass Company.

2. F.B. Leopold.
3. NEFCO.

B. Materials:
   1. Fabricate trough of premium grade fiberglass reinforced polyester resin, with fiberglass constituting a nominal 30 percent by weight.
   2. Glass reinforcement shall be random chopped-strand type with a minimum strand length of 1 inch, and adequate contact molding pressure to provide complete wet-out of the glass fibers.
   3. Minimum physical properties:
      a. Tensile strength: 14,000 pounds per square inch.
      b. Flexural strength: 25,000 pounds per square inch.
      c. Flexural modulus: 1.0 times 10^6 pounds per square inch.
      d. Trough resin and all accessories shall be suitable for the specific operating environment.
   4. All trough supports and hardware shall be Type 316 stainless steel including bolts, nuts washers, straps, etc.
   5. All materials used in the construction of the troughs and accessories shall be approved for use in potable water.

C. Design:
   1. Trough shall be laminated of fiberglass reinforced polyester resin to a minimum thickness of 1/4 inch.
   2. Inside surface of each trough shall have a smooth gel coat finish, color as selected by the Engineer. The outside surface of each trough shall be resin sealed with no exposed glass fiber.
   3. Color shall be molded in and an ultraviolet inhibitor shall be used.
   4. Troughs shall have flat bottoms and vertical sides, unless otherwise indicated on the Drawings.
   5. Top edges of trough shall be straight with no more than 1/8 inch deviation from a true plane.
   6. Longitudinal stiffening ribs shall be integrally molded on the outside of the troughs to ensure rigidity.
   7. Spacer rods:
      a. Sufficient plastic spacer rods shall be included to maintain a uniform width over the length of each trough.
      b. Spacer rods shall be spaced to prevent buckling and to provide maximum resistance to water loading on the sidewall of the trough.
   8. Support system shall allow 1 inch minimum adjustment of the trough, horizontally and vertically, and shall allow no greater than the unsupported span divided by 1,000 (L/1,000) upward deflection while the trough is empty and water outside the trough is to the weir edge. Maximum vertical deflection shall not exceed the unsupported span divided by 1,000 (L/1,000) downward while the trough is full and water outside the trough is below the trough bottom.
   9. Thickness of laminate at all supports such as saddles shall be a least 150 percent of the nominal thickness of the trough.
   10. Fiberglass adjustable weir plates shall be factory assembled to the troughs as indicated on the Drawings.

11. Design troughs to span distance between support systems, within stress and deflection limitation, the following loadings:
    a. Gravity load: Downward vertical loads shall include the weight of the trough and appurtenant attachments, such as weir plates and spreader bars, together with the weight of the water to fill the trough. Any additional loads, such as piping shall also be considered.
    b. Buoyant load: The buoyant load shall act vertically upward, its magnitude equal to the weight of displaced water (trough weight neglected). The line of action passes through the centroid of the submerged cross-sectional area.
    c. Lateral load: Loads acting against the trough side walls, specifically those induced by differential water levels on either side of the trough walls. The maximum possible differential, existing when the trough is empty and the tank is full, or when the trough is full and the tank is empty, shall be used when calculating deflection, fiber stress, etc. Include sloshing load due to seismic forces on both the trough and the support system.
    d. Thermal stresses: The troughs shall be designed to accommodate temperature induced stresses resulting from different coefficients of thermal expansion and contraction between the trough and tank/support materials.
    e. Torsional stability: The trough system shall be designed to resist torsional oscillations induced by the flow of water over trough edges. Any or all of the following trough stabilization techniques shall be considered:
        1) Trough to trough stabilization by trough-to-support truss bracing.
        2) Torsional stiffness.
        3) Support spacing and rigidity.
        4) Internal baffles and or flow straighteners.
12. Deflection under load: Maximum vertical deflection under full buoyant load or gravity load shall be less than or equal to L/1,000, where L is defined as the unsupported trough length inches. Under no circumstance shall the maximum vertical deflection, measured at mid-point between trough supports, exceed 3/16 inches.
13. Maximum trough side wall horizontal deflection under full lateral load shall be less than or equal to D/100, where D is defined as the trough depth in inches. Under no circumstances shall the maximum side wall deflection exceed 3/16 inches.
14. Trough bottom deflection (oil canning) under full buoyant load or gravity load shall be less than or equal to W/100, where W is defined as the trough width, in inches. Under no circumstances shall the maximum bottom deflection exceed 3/16 inches.
15. Fiber stress limitations: Supplemental to the deflection criteria, the troughs shall also be designed such that the maximum wall stress under the most severe loading conditions is less than or equal to 1,500 pounds per square inch. This stress criterion is equivalent to 7:1 safety factor (approximate) as applied to the tensile and flexural properties of contact molded troughs and launders.
16. Thermal expansion and contraction: The troughs shall be designed to accommodate a thermally induced expansion and contraction of 1/8 inch per 20 feet length of trough over temperature range of 10 degrees Fahrenheit to 100 degrees Fahrenheit, without exceeding the deflection or strain limitations set forth in the proceeding paragraphs.

17. Provide calculations sealed and signed by a registered professional engineer registered in the state Colorado for seismic sloshing loads and design of trough support system. All other calculations shall be sealed and signed by a registered professional engineer.
18. Trough manufacturer shall be responsible to design the troughs and all supporting members and accessories required for a complete installation.
19. Seismic design criteria for troughs, supports, and anchors shall be as specified in Section 01850. Design for wind loads as specified in Section 01850.

PART 3 EXECUTION

3.01 EXAMINATION

A. Verify that conditions are satisfactory for installation of products as specified in Section 01600 - Product Requirements.

3.02 ERECTION AND INSTALLATION, GENERAL

A. Install products where indicated on the Drawings in accordance with manufacturer's printed instructions.

3.03 WEIRS

A. Carefully install weirs, aligning and leveling to the elevations indicated on the Drawings.

B. Installation tolerances:
   1. V-notch weirs. In the completed installation:
      a. The variation in elevation between any 2 notches of the weir plate in a tank shall not exceed 1/8 inch.
      b. In a round tank, the variation from elevation between any one quadrant of the weir and that of any other quadrant shall not exceed 1/16 inch.

3.04 PARSHALL FLUME AND CUTTHROAT FLUME

A. Install by grouting into place.

B. Install covers so that it is straight, plumb, and rigid.

3.05 GRATING

A. General:
   1. Contractor to verify measurements in field to adjust for field conditions.
   2. Contractor to determine correct size and shape of required grating sections based on field dimensions prior to cutting grating to size.
   3. Grating sections cut to a smaller width shall be cut flush with a bearing bar such that the edge is solid without protruding bars.
   4. Install grating with no gaps between adjacent sections, and a maximum gap of 1/2-in. between the edge of grating and the structure wall.

5. Lock grating panels securely in place with hold-down fasteners and hardware as required by the manufacturer.

B. Molded grating:
   1. Install grating with molded surface face upward, and ground and resin-coated surface face down.
   2. Seams between adjacent grating sections that occur directly over the support beams shall be no closer than 4 inches from edge of beam.
   3. Sealing: All shop and field cuts of grating shall be coated with 2 vinyl ester resin coats, with the final resin coat containing wax additive in the amount necessary to achieve a full cure of the surface.
   4. Connect straight edge of adjacent grating sections together at the midpoint of any unsupported spans.

3.06 TANK COVER

A. Install the cover system in accordance with the approved installation instructions and anchorage details.

B. If any panels must be field-cut, cut ends shall be dressed as per the manufacturer's recommendations.

C. Erection shall proceed according to sequence shown on the approved drawings.

D. Starting at the end shown on the manufacturer's drawings, position and place cover panels in locations as shown. Contractor must obtain manufacture's written consent for field modification including but not limited to cuts, copes, and holes.

E. Fasten or anchor FRP cover deck panels into location as indicated on the Drawings.

3.07 FIELD QUALITY CONTROL

A. Provide field quality control for the Work of this Section as required by Section 01450 - Quality Control.

B. Provided and paid by Contractor:
   1. Manufacturer's services:
      a. Furnish manufacturer's representative to conduct jobsite training in proper storage, handling, and installation of products for personnel who will perform the work. Owner representative may attend training sessions.

C. Provided and paid by Owner:
   1. Special Inspections: As scheduled on the Drawings, and as required by authorities having jurisdiction over the Work.
   2. Structural Observations: As scheduled on the Drawings.

END OF SECTION

[AD8] Addendum No. 8 - August 2024

Coversheet for:

NAN, INC

Project: HILO WWTP REHABILITATION & REPLACEMENT PHASE 1
Location: HILO, HAWAII, 96720

ANCHOR SCHEDULE AS APPLICABLE (U.N.O. IN THE DETAILS):

1. ALL ANCHOR SYSTEMS USED WITH STRUCTURAL CONNECTIONS, LADDERS AND GUARDRAIL POST-MOUNTS TO BE TYPE 316 S.S. AND PROVIDED BY OTHERS.
2. FOR ALL STRUCTURAL CONNECTIONS (OTHER THAN GUARDRAIL POST-MOUNTS) THE CONTRACTOR TO DETERMINE ANCHOR TYPE BASED ON ACTUAL FIELD CONDITIONS.
3. FOR ANCHORING TO CONCRETE (CRACKED OR UNCRACKED), THE FOLLOWING ANCHOR SYSTEMS ARE RECOMMENDED:
   - 3.a. HILTI KWIK BOLT TZ2 EXPANSION ANCHORS:
     MINIMUM EMBEDMENT IN BASE MATERIAL TO BE AS FOLLOWS:
     - FOR 3/8"Ø ANCHORS USE 2" EMBEDMENT. RECOMMENDED FASTENER IS 3/8"Ø x 3-3/4" S.S. 316
     - FOR 1/2"Ø ANCHORS USE 3-1/4" EMBEDMENT. RECOMMENDED FASTENER IS 1/2"Ø x 5-1/2" S.S. 316
     - FOR 5/8"Ø ANCHORS USE 4" EMBEDMENT. RECOMMENDED FASTENER IS 5/8"Ø x 6" S.S. 316
   - 3.b. HILTI HIT-RE 500 V3 EPOXY SYSTEM WITH HAS-R 316 S.S. THREADED RODS:
     MINIMUM EMBEDMENT IN BASE MATERIAL TO BE AS FOLLOWS:
     - FOR 3/8"Ø ANCHORS USE 3-3/8" EMBEDMENT
     - FOR 1/2"Ø ANCHORS USE 4-1/2" EMBEDMENT
     - FOR 5/8"Ø ANCHORS USE 5-5/8" EMBEDMENT
   - 3.c. ANCHOR BOLTS USED IN CONJUNCTION WITH GUARDRAIL POST SIDE-MOUNTS AND THE (4) ANCHOR STANCHION BASE TO BE HILTI HIT-RE 500 V3 WITH 1/2"Ø 316 S.S. HAS-R THREADED RODS WITH 4-1/2" MIN. EMBEDMENT
   - 3.d. ANCHOR BOLTS USED IN CONJUNCTION WITH THE (2) ANCHOR STANCHION BASE FOR GUARDRAIL POST TOP MOUNT TO BE HILTI HIT-RE 500 V3 WITH 1/2"Ø 316 S.S. HAS-R THREADED RODS WITH 6-3/4" MIN. EMBEDMENT
4. FOR ANCHORING TO MASONRY WALLS OR FOR ANCHOR SIZES NOT LISTED ABOVE SEE NOTES ON DETAIL SHEETS
5. ANCHOR SUBSTITUTION IS ALLOWED ONLY WITH PRIOR APPROVAL FROM FIBERGRATE AND WHEN USING EQUAL OR BETTER ANCHORS

In addition to field connections shown, corrections of minor misfits and a reasonable amount of cutting is considered as a part of erection. Any error which prevents assembly by moderate use of cutting is to be reviewed, and approval of correction or change is to be received in writing before proceeding.

Failure to Comply will relieve Fibergrate Composite Structures, Inc. of Responsibility.

FOR AREAS OF CONCRETE THAT DICTATE THE SIZE AND SHAPE OF GRATING, FIBERGRATE RECOMMENDS THAT CONTRACTOR PROVIDE COORDINATES INSTEAD OF TAPE-MEASURED DIMENSIONS TO DETERMINE THE EXACT LOCATION OF THE CONCRETE. THIS IS TO ASSURE PROPER FIT-UP OF GRATINGS.

LADDER CAGE AND LADDER SAFETY SYSTEM NOTES (AS APPLICABLE):

FOR LADDERS 24 FEET OR MORE IN LENGTH OR WHERE THE DISTANCE FROM TOP OF LADDER TO A LOWER LEVEL (GROUND, FLOOR OR ROOF) EXCEEDS 24 FEET (PER OSHA AND ANSI 14.3)

FOR LADDERS 20 FEET OR MORE IN LENGTH OR WHERE THE DISTANCE FROM TOP OF LADDER TO A LOWER LEVEL (GROUND, FLOOR OR ROOF) EXCEEDS 20 FEET (PER CalOSHA)

1. U.N.O. ALL LADDERS MUST BE PROVIDED WITH A LADDER SAFETY SYSTEM
2. CAGES ARE NOT CONSIDERED LADDER SAFETY SYSTEMS AND CAN BE PROVIDED ONLY IF USED IN COMBINATION WITH A LADDER SAFETY SYSTEM
3. LADDER-UP DEVICES AT TOP OF LADDERS ARE NOT CONSIDERED LADDER SAFETY SYSTEMS
4. IF NOT PROVIDED BY FIBERGRATE, LADDER CAGES OR LADDER SAFETY SYSTEMS ARE TO BE PROVIDED BY OTHERS

Design Criteria:

Grating:
LL = 100 psf uniformly distributed or 300 lb/ft concentrated load
Max. deflection: 0.25" or L/120, whichever controls.

Grating is capable of carrying uniform distributed load of 100 pounds per square foot on simple span of 48inches without deflection more than 0.08 inches.

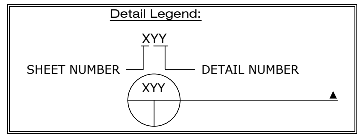
Maximum 0.11 inches at span of 48inches under concentrated load of 300 pounds.

Note:
Unless specifically noted, all grating and structural member selection is based on data published in Fibergrate's literature. Grating is not designed to resist motorized vehicular traffic.

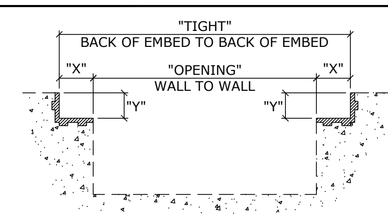
GENERAL NOTES (AS APPLICABLE)

1. ALL GRATING TO BE HELD DOWN WITH FIBERGRATE'S SS316 HOLD-DOWN CLIPS (HDCS); MIN. (2) HDCS PER SUPPORT BEAM (SPACED MAXIMUM (4) FEET ON CENTER), WITH MIN. (4) HDCS PER GRATING PANEL. OPEN MESH SINGLE-SPAN GRATING IN TRENCHES MAY BE INSTALLED WITHOUT HOLD DOWN CLIPS IF THE GRATING IS ADEQUATELY RESTRAINED TO PREVENT HORIZONTAL SLIDING, VERTICAL LIFTING, OR TIPPING. COVERED GRATING SHOULD BE HELD DOWN IN ALL CASES AND IN ALL SUPPORT CONDITIONS. IN CASE OF CONCRETE-EMBED-ANGLE SUPPORT, CONCRETE-ANCHORS/SCREWS MAY HAVE TO BE USED IN CONJUNCTION WITH HDCS.
2. FIBERGRATE'S SQUARE MESH MOLDED GRATING IS BI-DIRECTIONAL. FIBERGRATE'S RECTANGULAR MESH MOLDED GRATING IS TO BE TREATED AS UNI-DIRECTIONAL. FIBERGRATE'S PULTRUDED GRATING IS UNI-DIRECTIONAL.
3. CUT GRATING MAY HAVE OPEN STUB BARS.
4. ALL LOAD-BARS OF STAIR TREADS MUST BE FULLY SUPPORTED
5. ALL HORIZONTAL GUARDRAIL SHOWN IS 42" TALL FROM THE TOP OF WALKING SURFACE. FIBERGRATE'S STANDARD SQUARE TUBE GUARDRAIL SYSTEM CONSISTS OF 2-1/8" X 2-1/8" X 3/16" SQUARE TUBE POSTS, 1-3/4" X1-3/4"X1/8" SQUARE TUBE RAILS AND A 4" TALL KICKPLATE.
6. FIBERGRATE'S STANDARD FIXED LADDER SYSTEM CONSISTS OF 1-3/4" X 1-3/4" X 1/4" SQUARE TUBE RAILS AND 1-1/4" DIAMETER FLUTED RUNGS. STANDARD RUNGS ARE 18" CLEAR BETWEEN RAILS. SAFETY CAGES ARE PROVIDED WHEN REQUIRED.
7. THE RESPONSIBILITY FOR FULL COMPLIANCE WITH OSHA AND ANSI REGULATIONS FOR THE FINAL LADDER INSTALLATION LIES WITH THE CUSTOMER
8. ALL SHOP CUT EDGES AND DRILLED HOLES TO BE RESIN SEALED PRIOR TO SHIPPING.
9. SOME FIELD FABRICATION MAY BE REQUIRED, ALL FIELD CUT EDGES AND DRILLED HOLES TO BE SEALED USING SPRAY SEALER PROVIDED BY FIBERGRATE.
10. IT SHALL BE THE FIELD'S RESPONSIBILITY TO ASSURE PROPER FIT-UP OF F.R.P. ASSEMBLIES. THIS INCLUDES VERIFICATION OF AS-BUILT CONDITIONS. IN CASE OF CONCRETE MISALIGNMENT INCORRECTLY PLACED OR MISSING EMBEDS., THE FIELD SHALL MODIFY THE F.R.P. SUBASSEMBLIES AND PROVIDE THE REQUIRED ANCHORAGES.
11. CUTTING, FITTING, AND PLACEMENT: PERFORM CUTTING DRILLING, AND FITTING REQUIRED FOR INSTALLATION OF F.R.P. FABRICATIONS. SET F.R.P. FABRICATION ACCURATELY IN LOCATION ALIGNMENT, AND ELEVATION; WITH EDGES AND SURFACES LEVEL, PLUMB, TRUE, AND FREE OF RACK; AND MEASURED FROM ESTABLISHED LINES AND LEVELS.
12. ALL ANCHOR BOLTS TO BE PROVIDED (BY OTHERS) .
13. CONTRACTOR TO ENSURE THAT GRATING IS SUPPORTED AT INTERVALS AS DETERMINED FROM JOB SPECIFICATIONS AND FIBERGRATE'S PUBLISHED GRATING-LOAD-DEFLECTION CHARTS.
14. ALL DIMENSIONS TO BE VERIFIED BY CONTRACTOR. CONTRACTOR ALSO TO VERIFY DIMENSIONS WHERE NOTED.

Detail Legend:



NOTE: THIS DESIGN IS BASED ON MATERIALS MANUFACTURED BY AND/OR SUPPLIED SOLELY BY FIBERGRATE COMPOSITE STRUCTURES, INC. MATERIAL SUBSTITUTIONS OF ANY KIND WITHOUT THE EXPRESSED WRITTEN CONSENT OF FIBERGRATE COMPOSITE STRUCTURES, INC. RENDERS THE DESIGN INVALID AND NO LONGER CERTIFIED FOR THE DESIGN CONDITIONS STATED UNDER 'DESIGN CRITERIA'.



NOTE 1: WHEN PROVIDING TRENCH DIMENSIONS CUSTOMER WILL INDICATE TIGHT OR OPENING

NOTE 2: IF EMBED IS NOT PROVIDED BY FIBERGRATE, CUSTOMER WILL ALSO PROVIDE "X" & "Y" DIMENSIONS

TIGHT VS. OPENING DIMENSIONING

DOMESTIC MATERIALS QUOTED

SOME OR ALL MATERIALS WITHIN THIS PROJECT HAVE BEEN QUOTED AS DOMESTIC IN ORIGIN.

DRAWING APPROVAL

NAN, INC
CERTIFIES THAT FIBERGRATE COMPOSITE STRUCTURES INC. DRAWING #720006934, SHEET(S) 1 THROUGH 5 REV. 0 HAVE BEEN CAREFULLY REVIEWED AND HEREBY RELEASES THESE DRAWINGS FOR FABRICATION WITH THE FOLLOWING IN FORCE:

1) FIBERGRATE COMPOSITE STRUCTURES INC. ASSUMES FULL RESPONSIBILITY FOR THE ACCURATE TAKE-OFF OF QUANTITIES AND DIMENSIONS AS SHOWN IN CONTRACT OR BID DRAWINGS, PROVIDED TO FIBERGRATE COMPOSITE STRUCTURES INC., FROM WHICH THESE DRAWINGS WERE GENERATED, AND

2) NAN, INC
ASSUMES FULL RESPONSIBILITY FOR CORRECTNESS OF ACTUAL FIELD DIMENSIONS IN ACCORDANCE WITH EITHER THE CONTRACT DRAWINGS, OR WITH THE DIMENSIONS CORRECTED ON THIS DRAWING. ANY VARIATION RESULTING IN ADDITIONAL COST IS THE RESPONSIBILITY OF:
NAN, INC.

APPROVED FOR PRODUCTION

☐ MAKE REVISIONS AS NOTED AND PROCEED WITH PRODUCTION.

☐ PROCEED WITH PRODUCTION NO CHANGES REQUIRED.

(CONTRACTOR SIGNATURE) (DATE)

REVISE AND RESUBMIT

☐ REVISE AND RESUBMIT FOR APPROVAL WITH THE CHANGES AS NOTED.

(CONTRACTOR SIGNATURE) (DATE)

"These drawings are for approval only, and are not intended for use as fabrication drawings"

CONTRACTOR TO VERIFY ALL DIMENSIONS

LEGEND:

Y=1 1/2"x1 1/2"x1/4" EMBED ANGLE

(R) = REMOVABLE GRATING

NO.	BY	DATE	REVISION	CHKD. BY	DATE
3					
2					
1					
0	VTR	08/29/25	APPROVAL	TJ	08/29/25

FABRICATION TOLERANCES (U.N.O.)

	MOLDED GRATING	PULTRUDED GRATING	DYNAFORM SHAPES
LENGTH	+1/8,-5/16	±1/8	±1/8
WIDTH	+1/8,-5/16	±1/8	±1/8
HOLE LOC.	N/A	N/A	±1/16
CIRC. CUTS	±1/8	±1/8	N/A
OVER-CUT	Up to 1/8" deep	Up to 1/8" deep	≤1/16
SQUARE OF CUT	±1/8	±1/8	±2°
ANGULARITY	N/A	N/A	±1°

DO NOT STACK TOLERANCES

THIS DRAWING AND ALL DESIGN INFORMATION IS THE SOLE PROPERTY OF FIBERGRATE COMPOSITE STRUCTURES INC. DRAWING IS NOT TO BE DISCLOSED OR COPIED WITHOUT THE WRITTEN AUTHORIZATION OF FIBERGRATE COMPOSITE STRUCTURES INC.

GRATING: I5015-6 VEFR 1-1/2" DEEP PULTRUDED GRATING
COLOR: YELLOW SURFACE: GRIT SURFACE
TREADS:
COLOR: SURFACE:
DYNAFORM SHAPES: VEFR
COLOR: DK GRAY
DYNAFORM GUARDRAIL:
COLOR:
COATING:
DYNAFORM LADDER:
COLOR:
HARDWARE: 316 SS
GRATING H.D.C. ASSYS.: FC-2
REFERENCE DWGS.: PDF NO: 03-S-01-107 R0, 06-S-01-101 R0 & 09-S-01-101 R1

CONTRACTOR: NAN, INC
PROJECT: HILO WWTP REHABILITATION & REPLACEMENT PHASE 1 (#202300067461)
LOCATION: HILO, HAWAII, 96720
P.O. NO.: 82356
DRAWN BY: VTR
DATE: 08/29/25
STRUCTURAL DESIGNER: Raul Vazquez
PROJECT COORDINATOR: T. Truss

Fiberglass Grating ● Structurals ● Fabrication

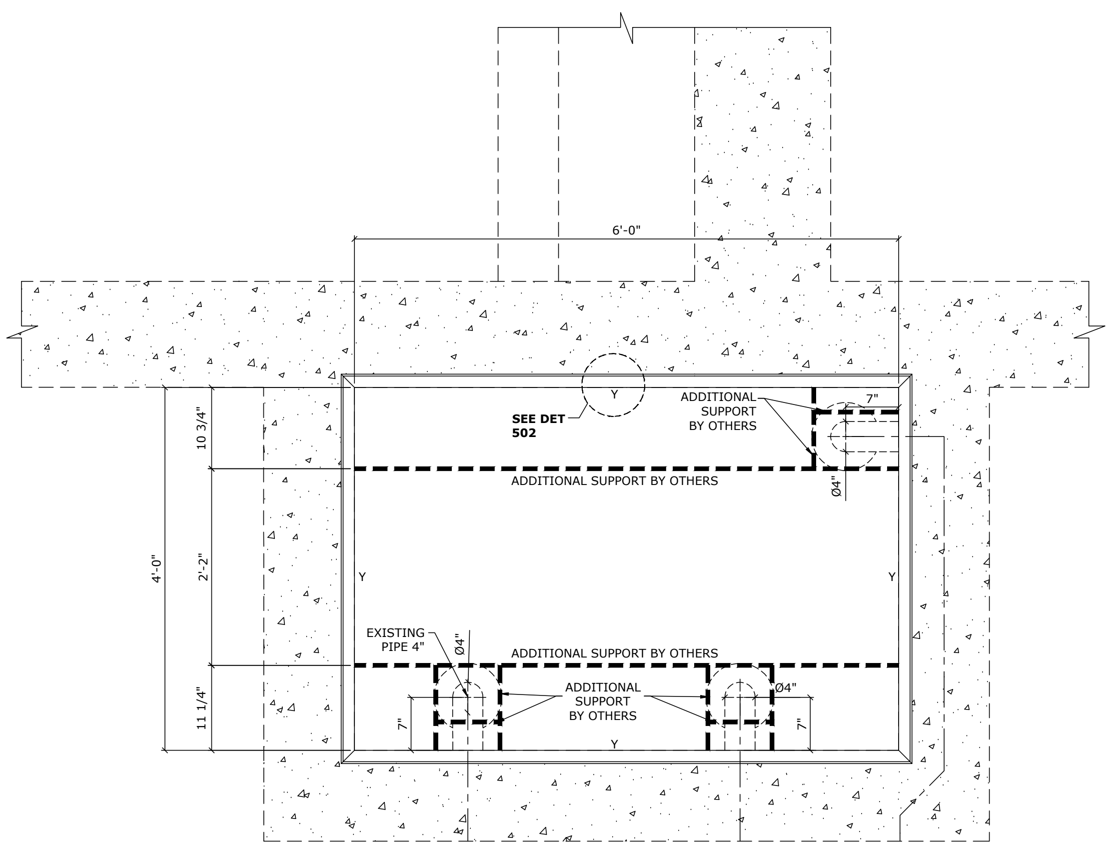
Fibergrate Composite Structures

900 FM 205 • STEPHENVILLE, TEXAS 76401 • Phone: (800) 527-4043

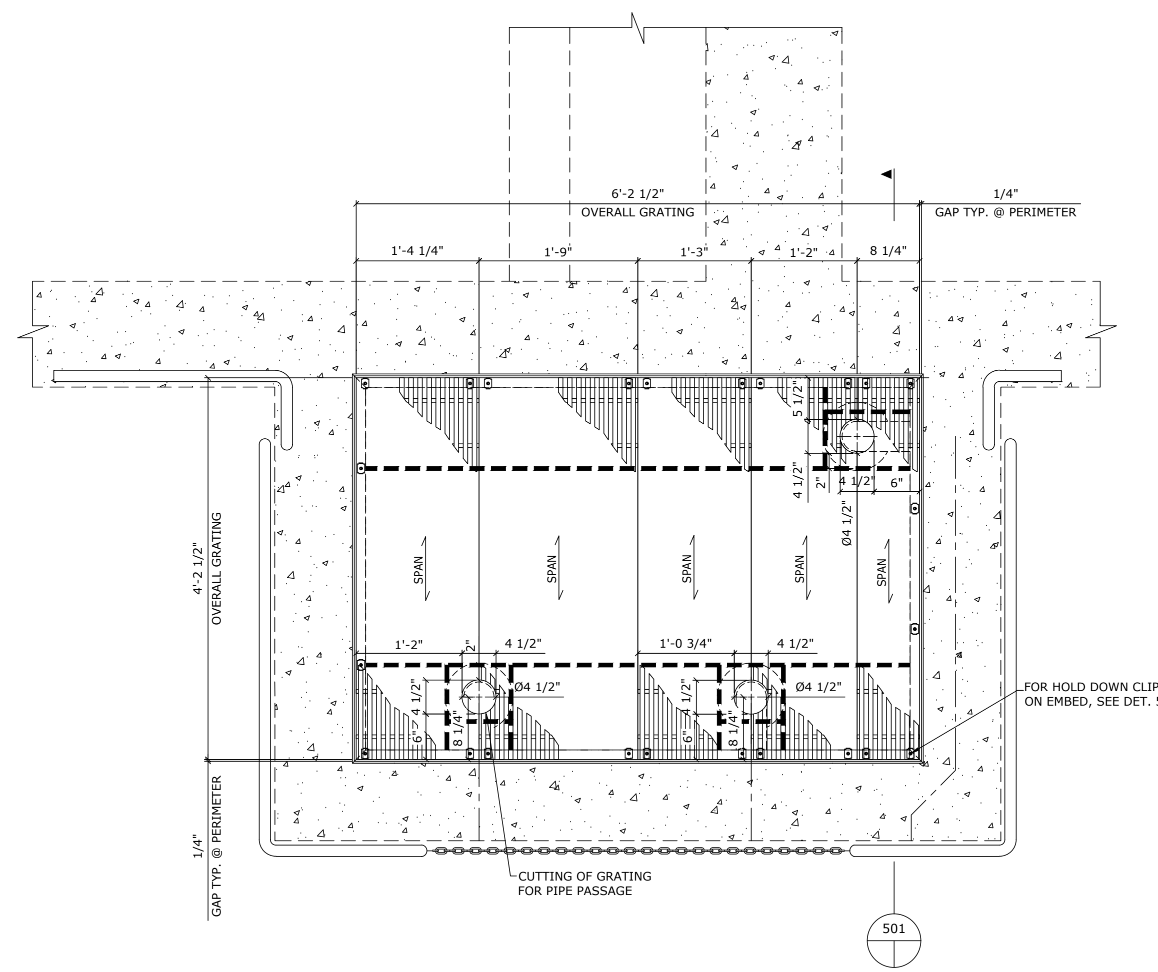
Cust NO.: BP0029538
W.O. NO.: 720006934
PROJECT NUMBER: PRJ018438
DWG. SHT.: 1 OF 5

0.0833

FEW DIMENSIONS WERE SCALED FROM CUSTOMER PROVIDED PDF'S, CONTRACTOR TO VERIFY ALL DIMENSIONS.



STRUCTURAL PLAN



GRATING PLAN

201 PRIMARY SEDIMENTATION TANKS

1 AREA REQUIRED

REF. SHEET(S) - 03-S-01-107-R0

"These drawings are for approval only, and are not intended for use as fabrication drawings"

CONTRACTOR TO VERIFY ALL DIMENSIONS

LEGAND:

LEGEND:

Y=1 1/2"x1 1/2"x1/4" EMBED ANGLE

(R) = REMOVABLE GRATING

NO.	BY	DATE	REVISION	CHKD. BY	DATE
3					
2					
1					
0	VTR	08/29/25	APPROVAL	TJ	08/29/25

FABRICATION TOLERANCES (U.N.O.)

	MOLDED GRATING	PULTRUDED GRATING	DYNAFORM SHAPES
LENGTH	+1/8,-5/16	±1/8	±1/8
WIDTH	+1/8,-5/16	±1/8	±1/8
HOLE LOC.	N/A	N/A	±1/16
CIRC. CUTS	±1/8	±1/8	N/A
OVER-CUT	Up to 1/8" deep	Up to 1/8" deep	≤1/16
SQUARE OF CUT	±1/8	±1/8	±2°
ANGULARITY	N/A	N/A	±1°

DO NOT STACK TOLERANCES

GRATING: I5015-6 VEFR 1-1/2" DEEP PULTRUDED GRATING

COLOR: YELLOW SURFACE: GRIT SURFACE

TREADS:

COLOR: SURFACE:

DYNAFORM SHAPES: VEFR

COLOR: DK GRAY

DYNAFORM GUARDRAIL:

COLOR:

COATING:

DYNAFORM LADDER:

COLOR:

HARDWARE: 316 SS

GRATING H.D.C. ASSYS.: FC-2

REFERENCE DWGS.: PDF NO: 03-S-01-107 R0

CONTRACTOR: NAN, INC

PROJECT: HILO WWTP REHABILITATION & REPLACEMENT PHASE 1 (#202300067461)

LOCATION: HILO, HAWAII, 96720

P.O. NO.: 82356

DRAWN BY: VTR

DATE: 08/29/25

STRUCTURAL DESIGNER: Raul Vazquez

PROJECT COORDINATOR: T. Truss

Fiberglass Grating ● Structurals ● Fabrication

Fibergrate Composite Structures

900 FM 205 • STEPHENVILLE, TEXAS 76401 • Phone: (800) 527-4043

Cust NO.: BP0029538

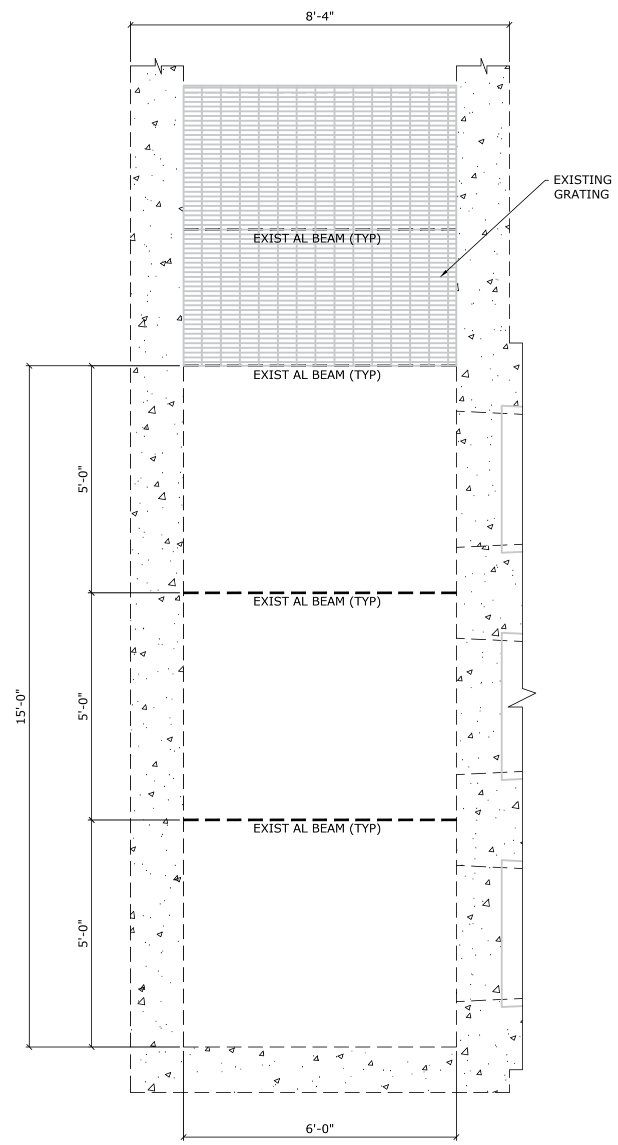
W.O. NO.: 720006934

PROJECT NUMBER: PRJ018438

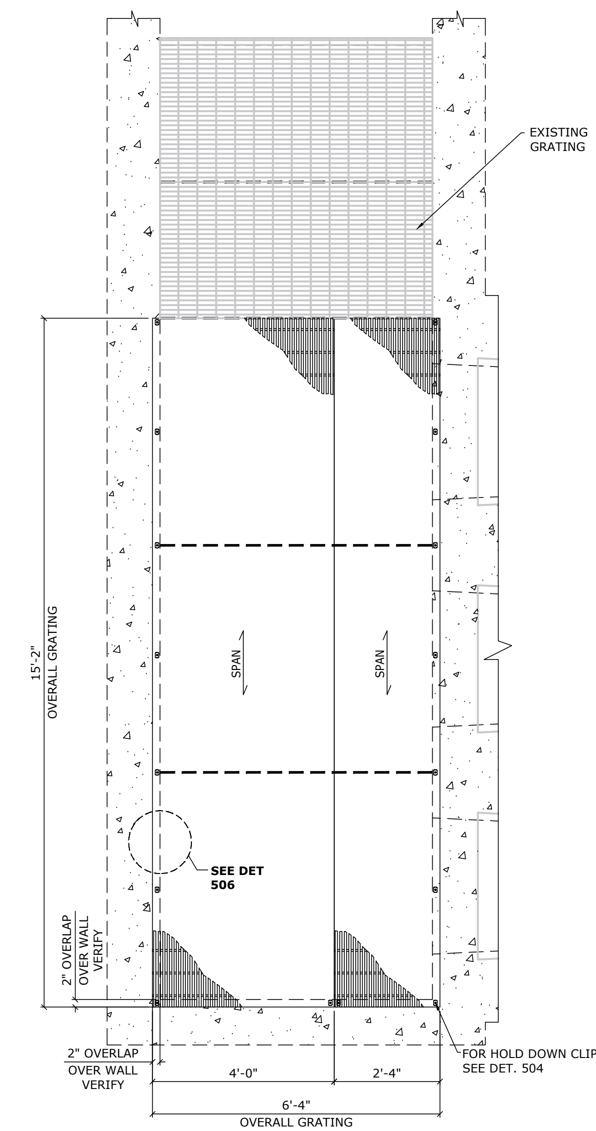
DWG. SHT.: 2 OF 5

0.0833

FEW DIMENSIONS WERE SCALED FROM CUSTOMER PROVIDED PDF'S, CONTTRACTOR TO VERIFY ALL DIMENSIONS.



STRUCTURAL PLAN



GRATING PLAN

301

SECONDARY CLARIFIER

1 AREA REQUIRED
REF. SHEET(S) - 06-S-01-101 R0

XX

"These drawings are for approval only, and are not intended for use as fabrication drawings"

CONTRACTOR TO VERIFY ALL DIMENSIONS

LEGEND:

Y=1 1/2"x1 1/2"x1/4" EMBED ANGLE

R = REMOVABLE GRATING

NO.	BY	DATE	REVISION	CHKD. BY	DATE
3					
2					
1					
0	VTR	08/29/25	APPROVAL	TJ	08/29/25

FABRICATION TOLERANCES (U.N.O.)

	MOLDED GRATING	PULTRUDED GRATING	DYNAFORM SHAPES
LENGTH	+1/8,-5/16	±1/8	±1/8
WIDTH	+1/8,-5/16	±1/8	±1/8
HOLE LOC.	N/A	N/A	±1/16
CIRC. CUTS	±1/8	±1/8	N/A
OVER-CUT	Up to 1/8" deep	Up to 1/8" deep	≤1/16
SQUARE OF CUT	±1/8	±1/8	±2°
ANGULARITY	N/A	N/A	±1°

DO NOT STACK TOLERANCES

GRATING: I5015-6 VEFR 1-1/2" DEEP PULTRUDED GRATING
COLOR: YELLOW SURFACE: GRIT SURFACE
TREADS:
COLOR: SURFACE:
DYNAFORM SHAPES: VEFR
COLOR: DK GRAY
DYNAFORM GUARDRAIL:
COLOR:
COATING:
DYNAFORM LADDER:
COLOR:
HARDWARE: 316 SS
GRATING H.D.C. ASSYS.: FC-2
REFERENCE DWGS.: PDF NO: 06-S-01-101 R0

CONTRACTOR: NAN, INC
PROJECT: HILO WWTP REHABILITATION & REPLACEMENT PHASE 1 (#202300067461)
LOCATION: HILO, HAWAII, 96720
P.O. NO.: 82356
DRAWN BY: VTR
DATE: 08/29/25
STRUCTURAL DESIGNER: Raul Vazquez
PROJECT COORDINATOR: T. Truss

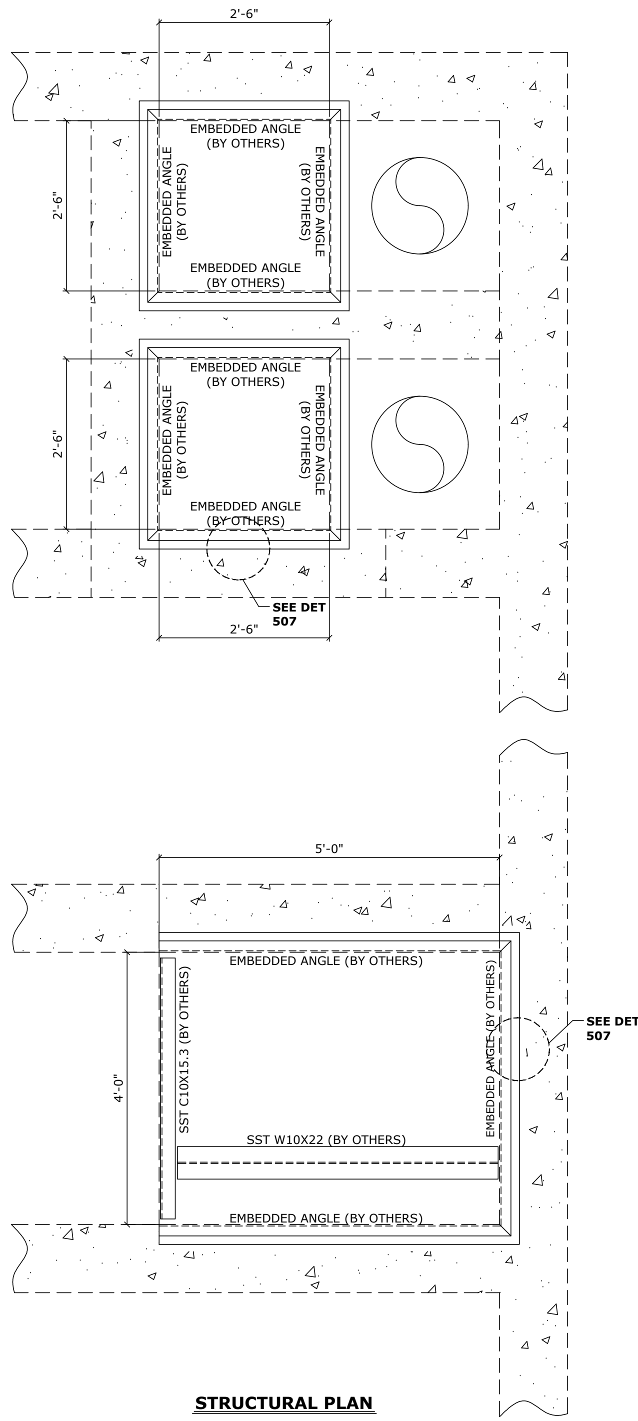
Fiberglass Grating ● Structurals ● Fabrication

Fibergrate Composite Structures

900 FM 205 • STEPHENVILLE, TEXAS 76401 • Phone: (800) 527-4043

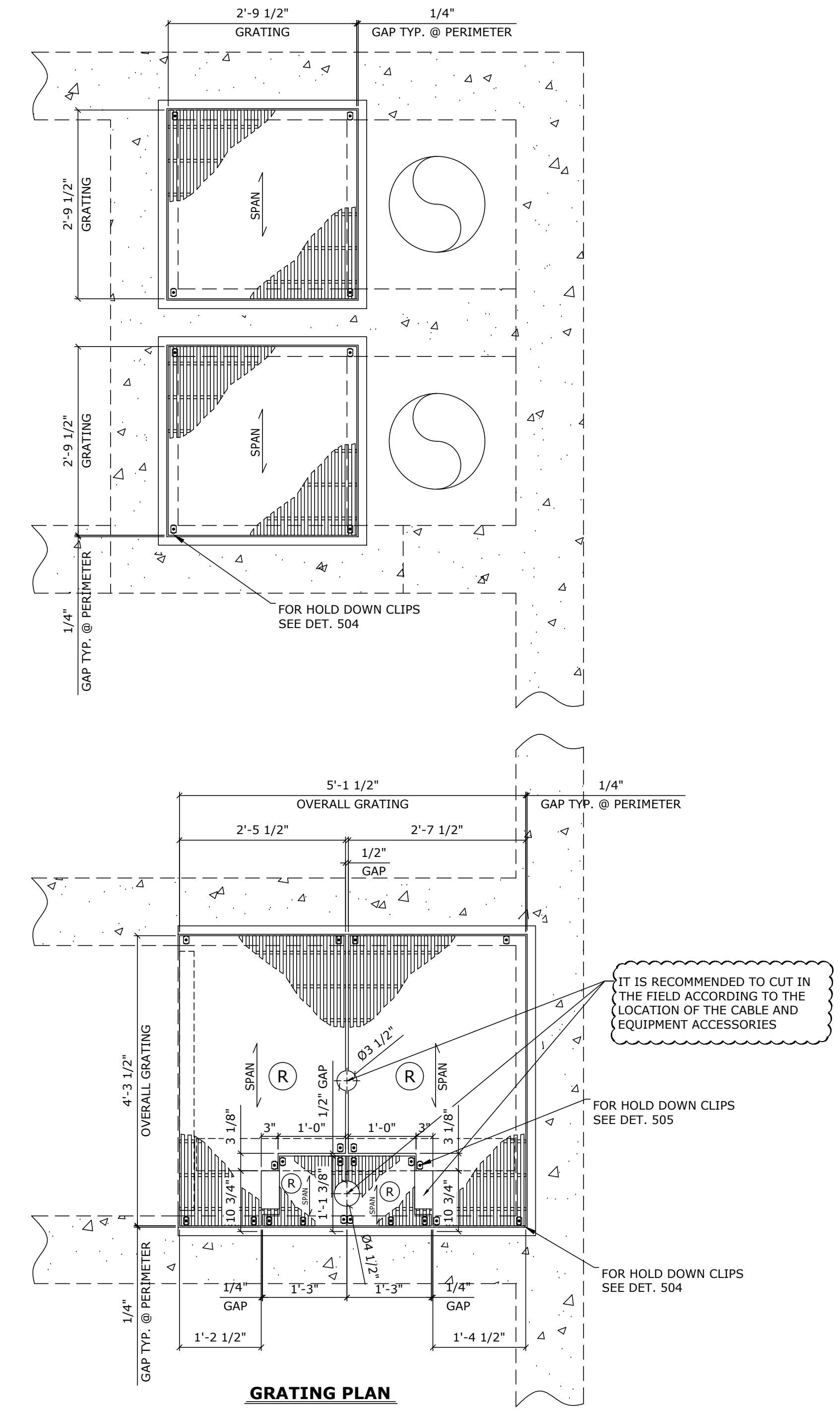
Cust NO.: BP0029538
W.O. NO.: 720006934
PROJECT NUMBER: PRJ018438
DWG. SHT.: 3 OF 5

0.0417



STRUCTURAL PLAN

FEW DIMENSIONS WERE SCALED FROM CUSTOMER PROVIDED PDF'S, CONTTRACTOR TO VERIFY ALL DIMENSIONS.



GRATING PLAN

401 — 3W PUMP STATION AND EFFLUENT FLUME
1 AREA REQUIRED
REF. SHEET(S) - 09-S-01-101 R1

"These drawings are for approval only, and are not intended for use as fabrication drawings"

CONTRACTOR TO VERIFY ALL DIMENSIONS

LEGEND:

Y=1 1/2"x1 1/2"x1/4" EMBED ANGLE

(R) = REMOVABLE GRATING

NO.	BY	DATE	REVISION	CHKD. BY	DATE
3					
2					
1					
0	VTR	08/29/25	APPROVAL	TJ	08/29/25

FABRICATION TOLERANCES (U.N.O.)

	MOLDED GRATING	PULTRUDED GRATING	DYNAFORM SHAPES
LENGTH	+1/8,-5/16	±1/8	±1/8
WIDTH	+1/8,-5/16	±1/8	±1/8
HOLE LOC.	N/A	N/A	±1/16
CIRC. CUTS	±1/8	±1/8	N/A
OVER-CUT	Up to 1/8" deep	Up to 1/8" deep	≤1/16
SQUARE OF CUT	±1/8	±1/8	±2°
ANGULARITY	N/A	N/A	±1°

DO NOT STACK TOLERANCES

GRATING: I5015-6 VEFR 1-1/2" DEEP PULTRUDED GRATING
COLOR: YELLOW SURFACE: GRIT SURFACE
TREADS:
COLOR: SURFACE:
DYNAFORM SHAPES: VEFR
COLOR: DK GRAY
DYNAFORM GUARDRAIL:
COLOR:
COATING:
DYNAFORM LADDER:
COLOR:
HARDWARE: 316 SS
GRATING H.D.C. ASSYS.: FC-2
REFERENCE DWGS.: PDF NO: 09-S-01-101 R1

CONTRACTOR: NAN, INC
PROJECT: HILO WWTP REHABILITATION & REPLACEMENT PHASE 1 (#202300067461)
LOCATION: HILO, HAWAII, 96720
P.O. NO.: 82356
DRAWN BY: VTR
DATE: 08/29/25
STRUCTURAL DESIGNER: Raul Vazquez
PROJECT COORDINATOR: T. Truss

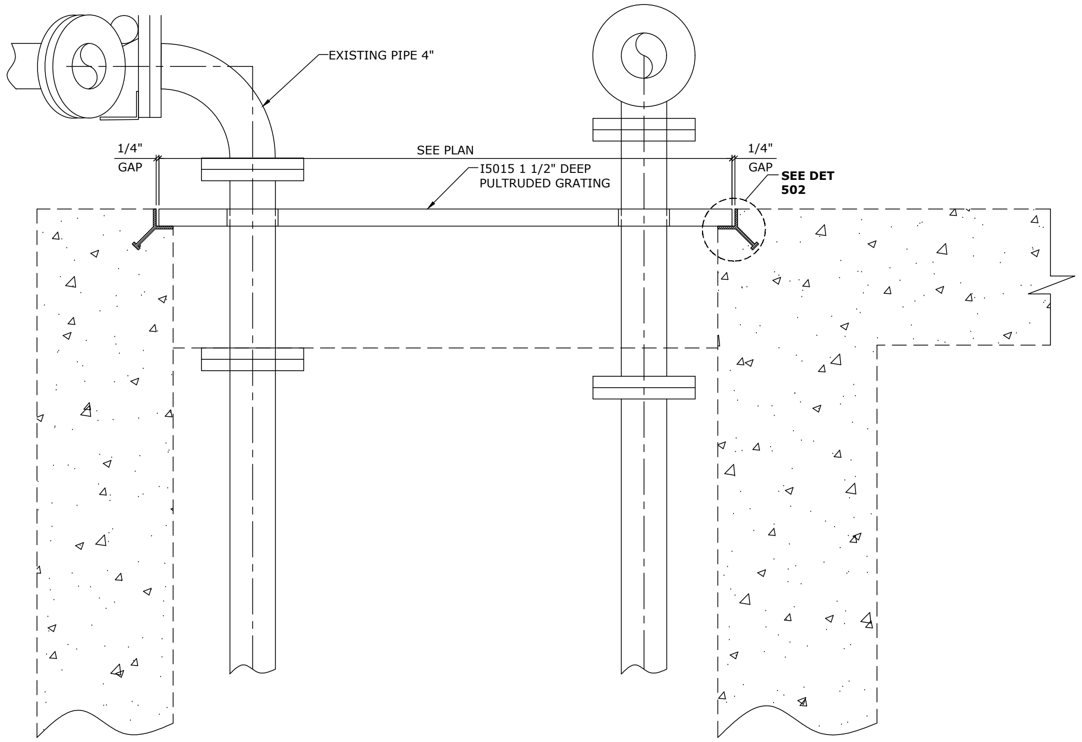
Fiberglass Grating ● Structurals ● Fabrication

Fibergrate Composite Structures

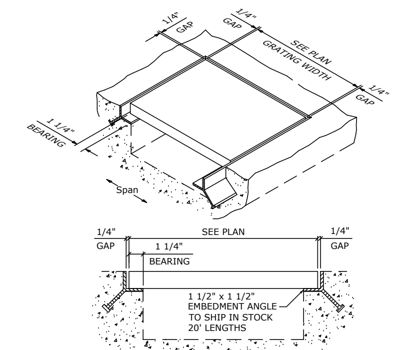
900 FM 205 • STEPHENVILLE, TEXAS 76401 • Phone: (800) 527-4043

Cust NO.: BP0029538
W.O. NO.: 720006934
PROJECT NUMBER: PRJ018438
DWG. SHT.: 4 OF 5

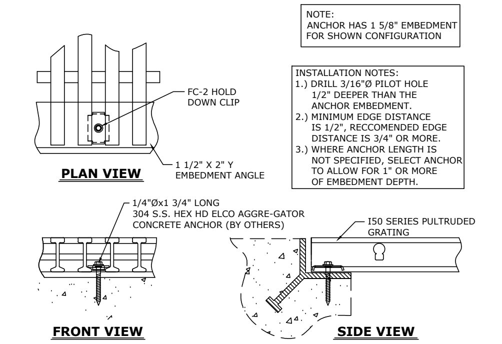
0.0625



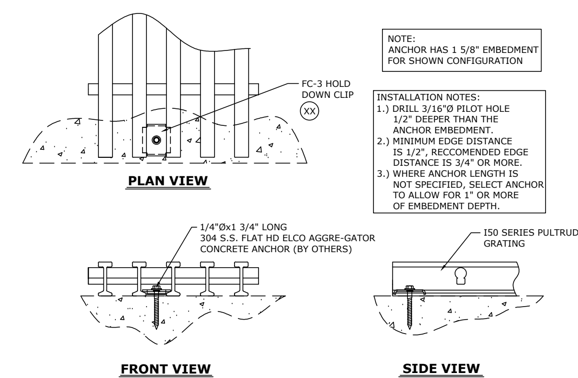
501 SECTION
1 AREA REQUIRED
REF. SHEET(S) - 03-M-01-305



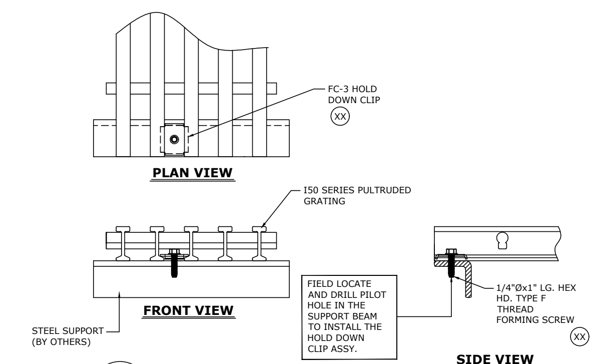
502 TRADITIONAL-EMBEDMENT ANGLE
SHOWN WITH 1 1/2" DEEP GRATING



503 FC-2 HOLD DOWN CLIP ASSEMBLY
SCALE-2X

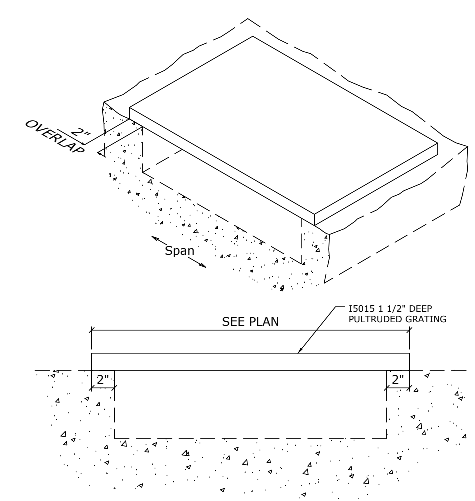


504 FC-2 HOLD DOWN CLIP TO CONCRETE ASSEMBLY
SCALE-2X

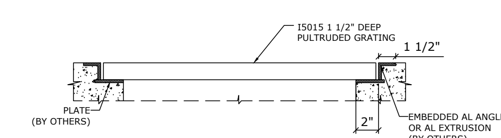


505 FC-2 HOLD DOWN CLIP ASSEMBLY
FOR REMOVABLE GRATING
SCALE-2x

FEW DIMENSIONS WERE SCALED FROM CUSTOMER PROVIDED PDF'S, CONTRACTOR TO VERIFY ALL DIMENSIONS.



506 GRATING OVERLAP
SHOWN WITH 1 1/2" DEEP GRATING



507 GRATING- REBATE AND SEAT
REF. SHEET(S) - 00-T-03-713 / Detail S542

XX

"These drawings are for approval only, and are not intended for use as fabrication drawings"

CONTRACTOR TO VERIFY ALL DIMENSIONS

LEGEND:

Y=1 1/2"x1 1/2"x1/4" EMBED ANGLE

R = REMOVABLE GRATING

NO.	BY	DATE	REVISION	CHKD. BY	DATE
3					
2					
1					
0	VTR	08/29/25	APPROVAL	TJ	08/29/25

FABRICATION TOLERANCES (U.N.O.)

	MOLDED GRATING	PULTRUDED GRATING	DYNAFORM SHAPES
LENGTH	+1/8,-5/16	±1/8	±1/8
WIDTH	+1/8,-5/16	±1/8	±1/8
HOLE LOC.	N/A	N/A	±1/16
CIRC. CUTS	±1/8	±1/8	N/A
OVER-CUT	Up to 1/8" deep	Up to 1/8" deep	≤1/16
SQUARE OF CUT	±1/8	±1/8	±2°
ANGULARITY	N/A	N/A	±1°

DO NOT STACK TOLERANCES

GRATING: I5015-6 VEFR 1-1/2" DEEP PULTRUDED GRATING
COLOR: YELLOW SURFACE: GRIT SURFACE
TREADS:
COLOR: SURFACE:
DYNAFORM SHAPES: VEFR
COLOR: DK GRAY
DYNAFORM GUARDRAIL:
COLOR:
COATING:
DYNAFORM LADDER:
COLOR:
HARDWARE: 316 SS
GRATING H.D.C. ASSYS.: FC-2
REFERENCE DWGS.: PDF DRAWINGS

CONTRACTOR: NAN, INC
PROJECT: HILO WWTP REHABILITATION & REPLACEMENT PHASE 1 (#202300067461)
LOCATION: HILO, HAWAII, 96720
P.O. NO.: 82356
DRAWN BY: VTR
DATE: 08/29/25
STRUCTURAL DESIGNER: Raul Vazquez
PROJECT COORDINATOR: T. Truss

Fiberglass Grating ● Structurals ● Fabrication

Fibergrate Composite Structures

900 FM 205 • STEPHENVILLE, TEXAS 76401 • Phone: (800) 527-4043

Cust NO.: BP0029538
W.O. NO.: 720006934
PROJECT NUMBER: PRJ018438
DWG. SHT.: 5 OF 5

0.1250